

CS101: Operating Systems & Kernel Concepts

Chapter 1: Operating Systems & Process Management

Operating system kernels manage system hardware, CPU scheduling, and process execution.

A process is an instance of a computer program being executed. Every process has a Process Control Block (PCB) containing process state, program counter, CPU registers, memory limits, and list of open files.

Process states include New, Ready, Running, Waiting, and Terminated.

Context switching is the mechanism of saving the CPU state of a running process into its PCB and restoring the state of another process to resume execution.

CPU scheduling algorithms include First-Come First-Served (FCFS), Shortest Job First (SJF), Round Robin (RR), and Priority Scheduling.

Chapter 2: Threads, Synchronization & Deadlocks

A thread is a lightweight unit of CPU utilization containing its own thread ID, program counter, register set, and stack. Multiple threads within the same process share code, data, and OS resources.

User-level threads are managed without OS kernel support, whereas kernel-level threads are managed directly by the operating system kernel.

Inter-Process Communication (IPC) allows processes to communicate via shared memory or message passing.

Concurrency issues arise when processes access shared data concurrently. Mutex locks and semaphores provide mutual exclusion.

A deadlock occurs when a set of processes are blocked because each process holds a resource and waits for another resource held by another process.

The four necessary conditions for deadlock are: Mutual Exclusion, Hold and Wait, No Preemption, and Circular Wait.

Chapter 3: Memory Management & Virtual Memory

Memory management keeps track of each byte of memory in the system and handles allocation to active processes.

Contiguous memory allocation assigns consecutive blocks of memory to processes using base and limit registers.

Paging is a memory management scheme that eliminates contiguous allocation requirements by dividing physical memory into fixed-size blocks called frames and logical memory into pages.

The Memory Management Unit (MMU) translates logical addresses into physical addresses using page tables.

Virtual memory is a memory management technique that provides an idealized abstraction of storage, allowing execution of processes that exceed physical memory size.

When a process accesses a page not currently in physical memory, a page fault exception occurs, triggering the OS to load the page from disk into RAM.

Page replacement algorithms include First-In First-Out (FIFO), Least Recently Used (LRU), and Optimal Page Replacement.

Chapter 4: File Systems & Storage Management

File systems provide a logical view of storage resources. File attributes include name, identifier, type, location, size, and protection permissions.

Directory structures organize files logically into single-level, two-level, or tree-structured directories.

Disk allocation methods include Contiguous Allocation, Linked Allocation, and Indexed Allocation (inode-based).

Disk scheduling algorithms optimize disk read/write head movement across tracks. Examples include FCFS, Shortest Seek Time First (SSTF), SCAN (Elevator Algorithm), and C-SCAN.